

LEARNER WORKBOOK

FULL NAME & SURNAME	
ID NUMBER:	
NAME OF ASSESSOR	
DATE OF ASSESSMENT	
VENUE	

MODEL ANSWERS

	ACHIEVED	NOT ACHIEVED
KNOWLEDGE		
SKILLS		

Signature of learner

Signature of Assessor

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This material has been compiled based on the **UNIT STANDARD 12484 PERFORM BASIC FIRE FIGHTING**

The program has been accredited by SASSETA in the Republic of South Africa.

Material compiled by: Training Dynamics Africa

HOW THIS LEARNER WORKBOOK IS COMPILED

This learner workbook has been compiled to guide you through the skills, values and knowledge reflected in this unit standard are required by people in the field of manufacturing and engineering. The learning outcomes in this unit standard also contribute to the exit level outcomes required for various manufacturing and engineering qualifications. Qualifying learners can select and use the appropriate fire fighting equipment to extinguish or control fires in the workplace.

1. Discuss and explain procedures for dealing with fires in the workplace.
2. identify the type of fire, its context and select the appropriate fire fighting procedure.
3. Identify, select and check appropriate fire fighting and safety equipment.
4. Fight containable / extinguishable fires.
5. Retreat from fire site and hand over to appropriate personnel.
6. Report / record status of fire and equipment.

You have to study the content and guidelines given in your learner reference guide to complete this workbook.

Remember that this completed workbook must be handed in as it will form part of your formative assessment and will play a role in awarding you the required credits that are to be achieved upon completion of this unit standard.

Write neatly and legibly and ensure that you do not simply copy the information contained in the learner reference guide into your workbook. Give your own opinion and ideas where applicable and write the answers in your own words to indicate your level of understanding and learning.

Good luck with the completion of this workbook.

LEARNING UNIT 1

Types of fires

LEARNING OUTCOMES

A learning credited with this unit will be able to:

1. Define a fire
2. Identify the elements of a fire
3. Demonstrate knowledge of the different types of fire
4. Identify the various causes of a fire

LEARNER ACTIVITY 1 (PAGE 13)**Define the word fire.**

It's a visible, tangible side effect of matter changing form: its one part of a chemical reaction.

Requirements achieved	YES		NO	
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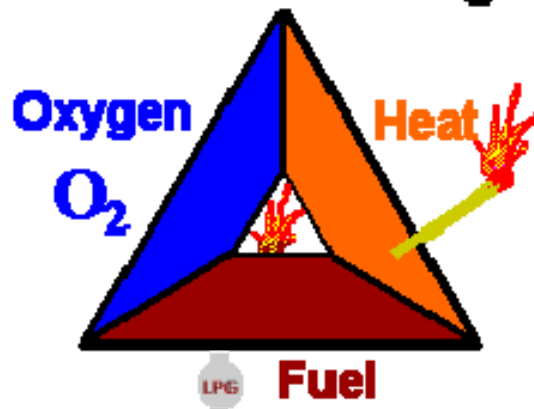
Comments in support of decision

LEARNER ACTIVITY 2 (PAGE 13)

Explain the elements of a fire in detail. Motivate your answer by drawing the fire triangle.

- ♦ Enough oxygen to sustain combustion,
- ♦ Enough heat to raise the material to its ignition temperature,
- ♦ Some sort of fuel or combustible material, and
- ♦ The chemical reaction that is fire.

The Fire Triangle



Make your drawing here:

Requirements achieved	YES		NO	
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Comments in support of decision

LEARNER ACTIVITY 3 (page 14)

Complete the attached diagram by filling in the missing blocks.

CLASSES OF FIRE		DESCRIPTION
A	 Common materials, such as paper, wood or most other combustibles.	Paper/trash/wood and other more-or-less organic solids. "Ordinary combustibles."
B	 Flammable liquids, such as gasoline paint remover or grease.	Inflammable liquids such as gasoline or paint thinner.
C	 Electrical fires.	Electrical fires, with electricity still flowing to the burning equipment.
D	 Combustible metals, usually found in industry.	Burning reactive metals, such as sodium, magnesium, titanium, and so forth. Such metals not only burn at high temperatures but can chemically extract oxygen from water and even carbon dioxide.

Requirements achieved	YES		NO	
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Comments in support of decision

LEARNER ACTIVITY 4(page 15)**List three causes of a fire.**

- ♦ **Careless smoking** is the leading cause of residential fire deaths. Smoke detectors and smolder-resistant bedding and upholstered furniture are significant fire deterrents.
- ♦ **Arson** is the second largest cause of residential fires and residential fire deaths. In commercial properties, arson is the major cause of deaths, injuries and dollar loss.
- ♦ **Heating** is the third leading cause of residential fire deaths. Heater fires are the leading cause of fire deaths.
- ♦ **Cooking** is the leading cause of apartment fires and the second most frequent cause of single-family residential fires. These fires often result from unattended cooking and human error, rather than from mechanical failures of stoves or ovens.
- ♦ **Wiring**, outlets, switches, circuit breakers and other electrical devices are the third leading cause of fires and the second leading cause of fire deaths.

Requirements achieved	YES		NO	
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Comments in support of decision

SELF-ASSESSMENT

The learner must make use of the following self-evaluation checklist to rate himself against the learning outcomes of this particular training module in establishing the level of mastery of the information.

1. Not able to comply
2. Reasonable compliance (Not acceptable for final evaluation)
3. Able to comply fully

LEARNING OUTCOMES		1	2	3
1	Define a fire			
2	Identify the elements of a fire			
3	Demonstrate knowledge of the different types of fire			
4	Identify the various causes of a fire			

Evidence Collector Signature

Date

Security Officer Signature

Date

LEARNING UNIT 2

Fire Prevention

LEARNING OUTCOMES

A learning credited with this unit will be able to:

1. Demonstrate knowledge of the goals of preventing a fire.
2. Demonstrate skills and knowledge to setup a strategy to prevent a fire.
3. List general measures to prevent a fire.

LEARNER ACTIVITY 1 (page 21)**List four goals to prevent a fire.**

- ♦ **Saving lives:** The primary goal of fire safety efforts is to protect building occupants from injury and to prevent loss of life.
- ♦ **Protecting property:** The secondary goal of fire safety is to prevent property damage.
- ♦ **Protecting operations:** By preventing fires and limiting damage we can assure that work operations will continue.

Requirements achieved	YES		NO	
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Comments in support of decision

LEARNER ACTIVITY 2 (page 22)**Explain the strategy of preventing a fire in detail.**

A fire must have three things to ignite and maintain combustion:

- ♦ Fuel
- ♦ Heat
- ♦ Oxygen

The basic strategy of fire prevention is to control or isolate sources of fuel and heat in order to prevent combustion.

If all three are not present in sufficient quantities a fire will not ignite or a fire will not be able to sustain combustion.

Requirements achieved	YES		NO	
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Comments in support of decision

LEARNER ACTIVITY 3 (page 23)**List four general measures to prevent a fire.**

- ♦ **Smoke Detector:** It is important to have smoke detectors installed on each floor.
- ♦ **Emergency Numbers:** While you should call emergency services if there is a fire, do not call from the location of the fire. You will be wasting valuable time you can be using to save your life.
- ♦ **Have a plan:** Make sure everyone is familiar with the evacuation plan and the location of the emergency doors. This is because if one exit route is blocked, you have a backup plan. Also, agree on a place to meet to do a “head count” once you evacuate the building.
- ♦ **Heaters:** Any portable heater should be kept at least a meter from anything that can burn (including the wall).
- ♦ **Matches:** It is very important to **keep matches and lighters out of a child’s reach**. Also, do not keep any near your fireplace.
- ♦ **Cooking:** Keep the cooking areas clear of combustibles. Make sure to keep all pot handles turned inward so they are not accidentally knocked over. Try to wear short sleeves or fitted clothes so your sleeves don’t catch fire.
- ♦ **Fireplace:** Before each new heating season, it is a good idea to have your chimney cleaned. Also, do not keep any matches, newspapers, or kindling near the fireplace.
- ♦ **Clothes Dryer:** Never leave your clothes dryer on when you are not home.
- ♦ **Electric appliances & wiring**

Requirements achieved	YES		NO	
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Comments in support of decision

SELF-ASSESSMENT

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1. Not able to comply
2. Reasonable compliance (Not acceptable for final evaluation)
3. Able to comply fully

LEARNING OUTCOMES		1	2	3
1	Demonstrate knowledge of the goals of preventing a fire			
2	Demonstrate skills and knowledge to setup a strategy to prevent a fire			
3	List general measures to prevent a fire			

Evidence Collector Signature

Date

Security Officer Signature

Date

LEARNING UNIT 3

Housekeeping

LEARNING OUTCOMES

A learning credited with this unit will be able to:

1. Demonstrate the importance of house keeping
2. Apply good housekeeping guidelines according to company SOP
3. Demonstrate knowledge of storage guidelines

LEARNER ACTIVITY 1 (page 31)**Explain the importance of good housekeeping.**

- ♦ To reduce amounts of flammable and combustible materials.
- ♦ To reduce ignition hazards.
- ♦ To ensure safe emergency evacuation of occupants.
- ♦ To allow for quick emergency response.

Requirements achieved	YES		NO	
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Comments in support of decision

LEARNER ACTIVITY 2 (page 31)

Analyze the following statement and record your views.

“THE ONLY WAY TO CHECK DEFICIENCY IN THE WORKPLACE IS TO APPLY GOOD HOUSEKEEPING RULES.”

- ♦ Work areas, aisles, walkways, stairways, and equipment should be kept clear of loose materials, trash, scraps, etc.
- ♦ Never block aisles, fire exits, emergency equipment, or alarm pull stations with equipment or materials.
- ♦ Avoid build up of combustible trash and waste such as paper, wood, cardboard, etc.
- ♦ Keep use and storage of flammables and combustibles to a minimum.
- ♦ Clean up all spills such as grease, oil, or water immediately. A delay could result in accidents.

[illegible]

Requirements achieved	YES		NO	
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Comments in support of decision

LEARNER ACTIVITY 3 (page 33)**Discuss at least three storage guidelines.**

- ♦ No storage is allowed in corridors and stairwells. A cluttered hallway could slow down emergency evacuation.
- ♦ Storage must not exceed plane of 18 inches below sprinkler heads or smoke detectors. Storage that breaks this plane may prevent sprinkler heads from fully covering room during a fire.
- ♦ All storage must be at least 3 ft from electrical panels. In some emergency situations it will be necessary to access these panels quickly.

This image shows a blank sheet of white paper with horizontal ruling lines. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

Requirements achieved	YES		NO	
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Comments in support of decision

LEARNER ACTIVITY 4(page 34)

Draw a table indicating at least four do's and don'ts concerning housekeeping.

DO'S	DON'T'S
Keep floors clean and clear of waste.	Use aisles and stairways as storage areas.
Keep workplaces adequately lighted.	Allow materials to build up on floors.
Keep light fixtures clean.	Block emergency exits, fire equipment or first aid stations with stored materials.
Inspect and clean machinery and tools regularly.	Store compressed gases near heat sources.
Clean up spills immediately.	

This image shows a blank sheet of white paper with horizontal ruling lines. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

Requirements achieved	YES		NO	
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Comments in support of decision

SELF-ASSESSMENT

The learner must make use of the following self-evaluation checklist to rate himself against the learning outcomes of this particular training module in establishing the level of mastery of the information.

1. Not able to comply
2. Reasonable compliance (Not acceptable for final evaluation)
3. Able to comply fully

LEARNING OUTCOMES		1	2	3
1	Demonstrate the importance of house keeping			
2	Apply good housekeeping guidelines according to company SOP			
3	Demonstrate knowledge of storage guidelines			

Evidence Collector Signature

Date

Security Officer Signature

Date

LEARNING UNIT 4

Fire fighting equipment

LEARNING OUTCOMES

A learning credited with this unit will be able to:

1. Know the different types of fire extinguishers
2. Know the use of the different types of fire extinguishers
3. Identify the correct fire extinguisher to fight a specific class of fire
4. How to operate a fire extinguisher
5. Know the location and operation of a fire hose

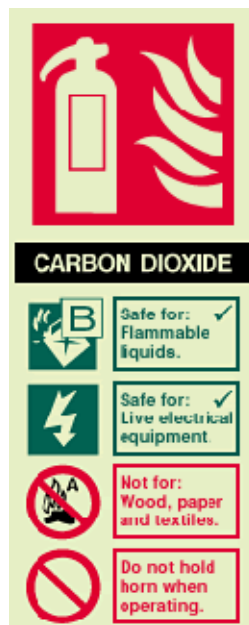
LEARNER ACTIVITY 1 (page 40)

You and your friend are having coffee when the coffee machine in cafeteria suddenly blows up.

There is a lot of electronic equipment all around the coffee machine. The flames are spreading very rapidly.

- ♦ What fire extinguisher will you use to extinguish the fire?
- ♦ How will you know that the fire extinguisher is the correct one?

Class C: Fires involve electrical equipment, such as appliances, wiring, circuit breakers and outlets. Class C extinguishers do not have a numerical rating. **The C classification means the extinguishing agent is non-conductive**



This image shows a single sheet of white paper with horizontal blue or grey ruling lines. The lines are evenly spaced and run across the width of the page. There are approximately 20 lines visible. The paper has a slight shadow on the right side, suggesting it's resting on a surface. The top edge of the paper is slightly irregular, possibly from being torn from a notebook.

Requirements achieved	YES		NO	
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Comments in support of decision

LEARNER ACTIVITY 2 (page 42)

Explain the correct use of the following fire extinguishers.

- ♦ **Water fire extinguisher.**
 - ♦ **Dry chemical extinguisher.**
 - ♦ **Carbon dioxide.**
- ♦ **Water extinguishers** or APW extinguishers (air-pressurized water) are suitable for **Class A fires ONLY**. Water extinguishers are filled with water and pressurized with oxygen.

Never use a water extinguisher on grease fires, electrical fires or class D fires. The flames will spread and make the fire bigger!

- ♦ **Dry chemical** extinguishers come in a variety of types and are suitable for a combination of **Class A, B and C fires**.

These are filled with foam or powder and pressurized with nitrogen.

BC: This is the regular type of dry chemical extinguisher. It is filled with sodium bicarbonate or potassium bicarbonate. The BC variety leaves a mildly corrosive residue which must be cleaned immediately to prevent any damage to materials.

ABC: This is the multipurpose dry chemical extinguisher. The ABC type is filled with mono-ammonium phosphate, a yellow powder that leaves a sticky residue that may be damaging to electrical appliances such as a computer.

- ♦ **Carbon Dioxide (CO₂) extinguishers** are used for **Class B and C fires**.

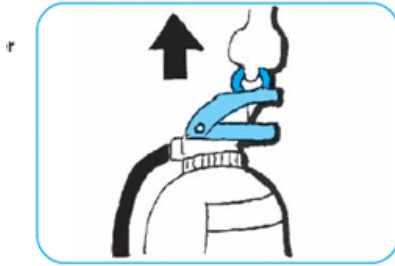
CO₂ extinguishers contain carbon dioxide, a non-flammable gas, and are highly pressurized. The pressure is so great that it is not uncommon for bits of dry ice to shoot out the nozzle.

This image shows a blank sheet of white paper with horizontal ruling lines. The lines are evenly spaced and extend across the width of the page. There are no margins, text, or other markings on the paper.

Requirements achieved	YES	NO
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Comments in support of decision

"P" stands for _____



This will unlock the operating handle and allow you to discharge the extinguisher.

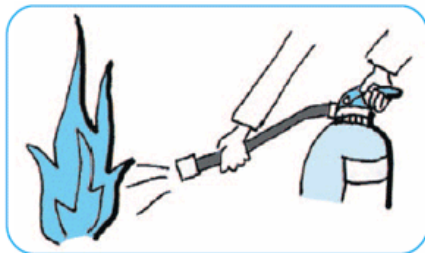
"A" stands for AIM



.....at the base of the fire.

This is important - in order to put out the fire, you must extinguish the fuel.

"S" stands for squeeze



This will release the extinguishing agent in the extinguisher. If the handle is released, the discharge will stop.

"S" stands for sweep



Using a sweeping motion, move the fire extinguisher back and forth until the fire is completely out.

LEARNER ACTIVITY 4 (page 49)**Explain the correct procedure to use a fire hose.**

- ♦ Ensure the nozzle or jet is in the closed position
- ♦ Turn on the main valve (some will not let the nozzle out until this is done)
- ♦ Pull the hose off the drum, towards the fire
- ♦ Open the nozzle or valve and direct the stream of water at the fire

Requirements achieved	YES		NO	
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Comments in support of decision

SELF-ASSESSMENT

The learner must make use of the following self-evaluation checklist to rate himself against the learning outcomes of this particular training module in establishing the level of mastery of the information.

1. Not able to comply
2. Reasonable compliance (Not acceptable for final evaluation)
3. Able to comply fully

LEARNING OUTCOMES		1	2	3
1	Know the different types of fire extinguishers			
2	Know the use of the different types of fire extinguishers			
3	Identify the correct fire extinguisher to fight a specific class of fire			
4	How to operate a fire extinguisher			
5	Know the location and operation of a fire hose			

Evidence Collector Signature

Date

Security Officer Signature

Date

LEARNING UNIT 5

Fighting Fire

LEARNING OUTCOMES

A learning credited with this unit will be able to:

1. Make the correct decision in the process of fighting a fire
2. Stop a fire by using the correct equipment and procedure
3. Apply general precautions when fighting a fire
4. Identify and apply the correct procedures after a fire has been put out accordingly to SOP

LEARNER ACTIVITY 1 (page 53)**What is the single most important factor to remember when fighting a fire?**

When a fire breaks out, you have only seconds to react, and in those few seconds, you must respond quickly and efficiently. You must stay calm and know your options. Frequently, the best option will simply be to leave the area and call for help. Yet, there may be other times when you will be in a position to put out the fire through the proper use of the right type of fire extinguisher.

Requirements achieved	YES		NO	
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Comments in support of decision

LEARNER ACTIVITY 2 (page 55)**Explain and discuss the general procedures when fighting a fire.**

Always keep the exit to your back while fighting a fire, and if your efforts don't seem to be successful, leave the area, closing doors behind you to assist in keeping the fire from spreading.

Once you are safe, call your local fire Department immediately.

Requirements achieved	YES		NO	
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Comments in support of decision

LEARNER ACTIVITY 3 (page 56)**Explain the term “REACT”.****REACT upon discovery of fire or smoke**

- ♦ **Remove** persons in immediate danger!
- ♦ **Ensure** doors are closed! (*confine fire/smoke*)
- ♦ **Activate** the building alarm!
- ♦ **Call** the Fire Department!
- ♦ **Treat** ALL fires as DANGEROUS!

Requirements achieved	YES		NO	
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Comments in support of decision

LEARNER ACTIVITY 4 (page 57)**What needs to be done after the fire has been put out?**

- ♦ WATCH the fire area
- ♦ If the fire flares up again repeat!
- ♦ If you can't control the fire, LEAVE immediately!
- ♦ Call the fire department to inspect the scene!
- ♦ Recharge or replace any used fire extinguisher!

Requirements achieved	YES		NO	
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Comments in support of decision

SELF-ASSESSMENT

The learner must make use of the following self-evaluation checklist to rate himself against the learning outcomes of this particular training module in establishing the level of mastery of the information.

1. Not able to comply
2. Reasonable compliance (Not acceptable for final evaluation)
3. Able to comply fully

LEARNING OUTCOMES		1	2	3
1	Make the correct decision in the process of fighting a fire			
2	Stop a fire by using the correct equipment and procedure			
3	Apply general precautions when fighting a fire			
4	Identify and apply the correct procedures after a fire has been put out accordingly to SOP			

Evidence Collector Signature

Date

Security Officer Signature

Date

LEARNING UNIT 6

Maintenance

LEARNING OUTCOMES

A learning credited with this unit will be able to:

1. Apply basic maintenance on fire fighting equipment

LEARNER ACTIVITY 1 (page 61)

Explain the importance of maintenance with specific reference to a fire extinguisher.

- ♦ The extinguisher is not blocked by equipment, coats or other objects that could interfere with access in an emergency.
- ♦ The pressure is at the recommended level. On extinguishers equipped with a gauge (such as that shown on the right), the needle should be in the green zone - not too high and not too low.
- ♦ The nozzle or other parts are not hindered in any way.
- ♦ The pin and tamper seal (if it has one) are intact.
- ♦ There are no dents, leaks, rust, chemical deposits and/or other signs of abuse/wear.
- ♦ Wipe off any corrosive chemicals, oil, gunk etc. that may have deposited on the extinguisher.

Requirements achieved	YES		NO	
Comments in support of decision				

SELF-ASSESSMENT

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1. Not able to comply
2. Reasonable compliance (Not acceptable for final evaluation)
3. Able to comply fully

LEARNING OUTCOMES		1	2	3
1	Apply basic maintenance on fire fighting equipment			

Evidence Collector Signature

Date

Security Officer Signature

Date